

EXPERIMENT NO 12

Determination and plotting of magnetic and electric field intensity due to infinite long conductor at any specific point located at H

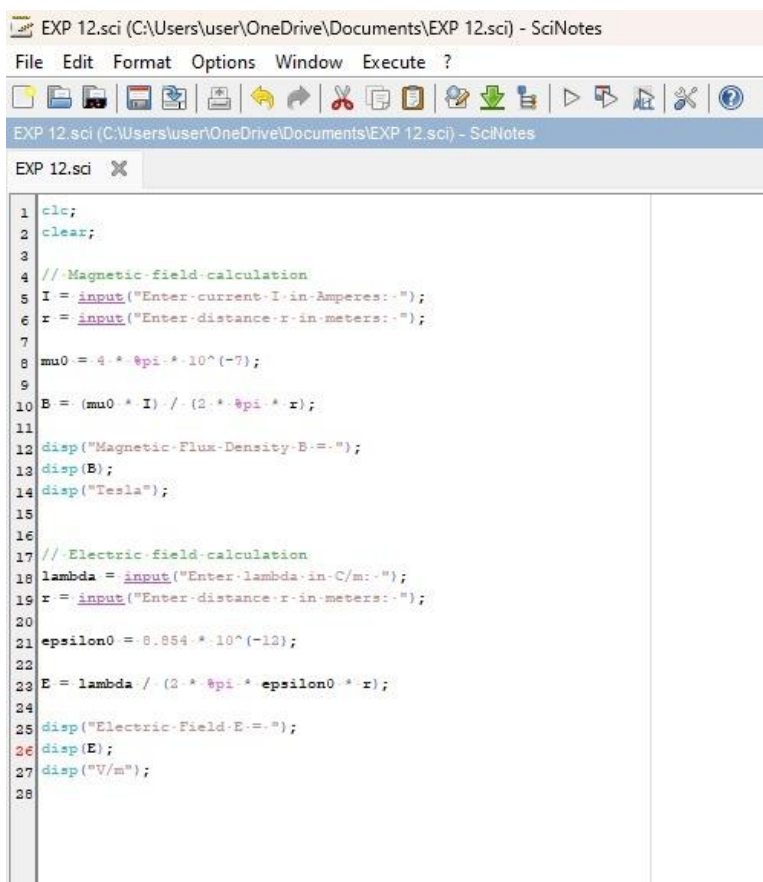
Aim:

To write program for Magnetic field intensity of infinite long conductor using SCILAB.

Software required:

- SCILAB version 6.0.1

Program:



```
EXP 12.sci (C:\Users\user\OneDrive\Documents\EXP 12.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 12.sci (C:\Users\user\OneDrive\Documents\EXP 12.sci) - SciNotes
EXP 12.sci X
1 clc;
2 clear;
3
4 //Magnetic-field-calculation
5 I = input("Enter current-I in-Amperes:-");
6 r = input("Enter distance-r in-meters:-");
7
8 mu0 = 4.*%pi.*10^(-7);
9
10 B = (mu0 * I) / (2.*%pi.*r);
11
12 disp("Magnetic-Flux-Density-B=-");
13 disp(B);
14 disp("Tesla");
15
16
17 //Electric-field-calculation
18 lambda = input("Enter-lambda-in-C/m:-");
19 r = input("Enter distance-r in-meters:-");
20
21 epsilon0 = 8.854.*10^(-12);
22
23 E = lambda / (2.*%pi.*epsilon0.*r);
24
25 disp("Electric-Field-E=-");
26 disp(E);
27 disp("V/m");
28
```

Output:

Scilab 2026.1.0 Console

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File Browser

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Documents

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- ex 6.scd
- EXP 12.scd

File/directory filter: Aa * ?

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Enter current I in Amperes: 5

Enter distance r in meters: 3

"Magnetic Flux Density B = "

0.0000003

"Tesla"

Enter lambda in C/m: 4

Enter distance r in meters: 5

"Electric Field E = "

1.438D+10

"V/m"

-->

Variable Browser

Name	Value	Type	Visibility	Memory
B	3.33e-07	Double	local	216 B
E	1.44e+10	Double	local	216 B
I	5	Double	local	216 B
epsilon0	8.85e-12	Double	local	216 B
lambda	4	Double	local	216 B
mu0	1.26e-06	Double	local	216 B
r	5	Double	local	216 B

Command History

```
// -- 10/08/2026 14:50:34 -- //  
// -- 24/08/2026 04:27:36 -- //  
// -- 24/08/2026 04:29:27 -- //  
-3*10^-7  
-6  
-4  
-2  
-5  
-3  
-4  
-5  
-3  
-4  
-5
```

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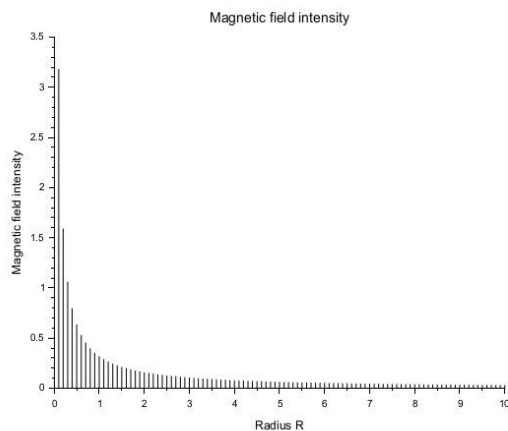
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Graphic window number 1

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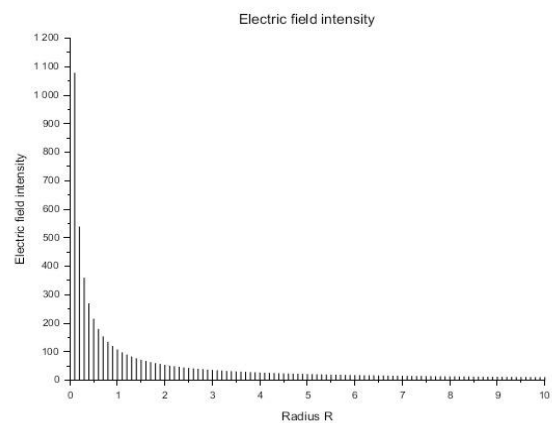
Graphic window number 1



Graphic window number 2

File Tools Edit ?

Graphic window number 2



Theoretical calculation:

EXP NO : 12

Determination and plotting of magnetic and electric field intensity due to infinite long conductor at any specific point located at H.

Output:

Enter I value = 5 A

Enter r value = 3 cm

Enter lambda value = 44 m

Enter z value = 5 cm

Practical output:

"Magnetic flux density $B = 0.0000003$ Tesla"

"Electric Field $E = 1.23 \times 10^6$ V/m"

Theoretical Calculation:

Given, current = 5 A

Distance $r = 3 \text{ cm} = 0.03 \text{ m}$

Line charge density $\lambda = 44 \text{ C/m} = 44 \times 10^{-6} \text{ C/m}$

$z = 5 \text{ cm} = 0.05 \text{ m}$

Magnetic flux Density

$$B = \frac{\mu_0 I}{2\pi r} \quad \mu_0 = 4\pi \times 10^{-7} \text{ H/m}$$

$$B = \frac{4\pi \times 10^{-7} \times 5}{2\pi \times 0.03} = 3.33 \times 10^{-5} \text{ T} = 0.0000003 \text{ T}$$

Electric Field intensity

$$E = \frac{\lambda}{2\pi\epsilon_0 R}$$

$$\text{where } R = \sqrt{r^2 + z^2} = \sqrt{(0.003)^2 + (0.05)^2} = 0.0583 \text{ m}$$
$$\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$$

$$\text{Therefore, } E = \frac{4 \times 10^{-6}}{2\pi (8.854 \times 10^{-12}) (0.0583)}$$

$$E = 1.23 \times 10^6 \text{ V/m}$$

$$B = 3.33 \times 10^{-5} \text{ T}$$

$$E = 1.23 \times 10^6 \text{ V/m}$$

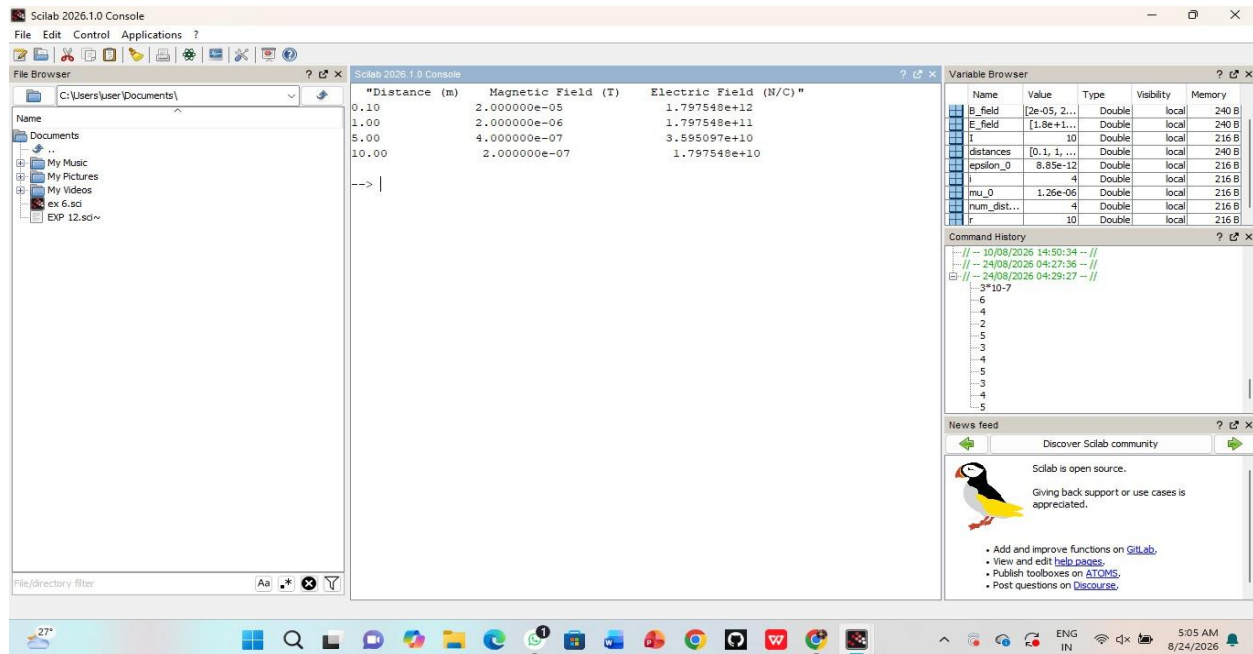
Case 1: To examine how the magnetic and electric field intensities change with varying distances from an infinitely long current-carrying conductor.

Program:

```
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File Edit Format Options Window Execute ?
EXP 12.sci (C:\Users\user\OneDrive\Documents\EXP 12.sci) - SciNotes
EXP 12.sci X

1  clc;
2  clear;
3
4  // Constants
5  mu_0 = 4 * pi * 10^(-7); // Permeability of free space
6  epsilon_0 = 8.854 * 10^(-12); // Permittivity of free space
7  I = 10; // Current in Amperes
8
9  // Distances from the conductor
10 distances = [0.1, 1, 5, 10]; // in meters
11 num_distances = length(distances);
12
13 // Initialise arrays
14 B_field = zeros(1, num_distances);
15 E_field = zeros(1, num_distances);
16
17 // Calculate magnetic and electric fields
18 for i = 1:num_distances
19     r = distances(i);
20
21     // Magnetic flux density
22     B_field(i) = (mu_0 * I) / (2 * pi * r);
23
24     // Electric field
25     E_field(i) = I / (2 * pi * epsilon_0 * r);
26 end
27
28 // Display results
29 disp("Distance (m) --- Magnetic Field (T) --- Electric Field (N/C)");
30
31 for i = 1:num_distances
32     fprintf("%0.2f --- %0.2f --- %0.2f\n", distances(i), B_field(i), E_field(i));
33 end
34
35
```

Output:



Case 2: To study how the magnetic and electric field intensities change for an infinitely long conductor at a fixed distance H but with varying currents.

Program:

```
EXP 12.sci (C:\Users\user\OneDrive\Documents\EXP 12.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 12.sci (C:\Users\user\OneDrive\Documents\EXP 12.sci) - SciNotes
EXP 12.sci
1 clear;
2 clear;
3
4 // Constants
5 H = 2; // Distance from conductor (m)
6 epsilon_0 = 8.854e-12; // Permittivity of free space (F/m)
7 lambda = 1.0e-6; // Linear charge density (C/m)
8
9 // Currents to evaluate
10 currents = [5, 10, 20, 30, 50];
11 num_currents = length(currents);
12
13 // Preallocate arrays
14 magnetic_field_intensity = zeros(1, num_currents);
15 electric_field_intensity = zeros(1, num_currents);
16
17 // Calculate magnetic and electric field intensities
18 for i = 1:num_currents
19     I = currents(i);
20
21     // Magnetic field intensity
22     magnetic_field_intensity(i) = I / (2 * pi * H);
23
24     // Electric field intensity
25     electric_field_intensity(i) = lambda / (2 * pi * epsilon_0 * H);
26
27 end
28
29 // Display results
30 disp("Current (A) --- Magnetic Field H (A/m) --- Electric Field E (N/C)");
31
32 for i = 1:num_currents
33     fprintf("%8.2f --- %8.6f --- %8.6e\n", ...
34         currents(i), ...
35         magnetic_field_intensity(i), ...
36         electric_field_intensity(i));
37 end
38
39
```

Output:

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EXP 12.sci

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Current (A)	Magnetic Field H (A/m)	Electric Field E (N/C)
5.00	0.397887	8.987742e+03
10.00	0.795775	8.987742e+03
20.00	1.591549	8.987742e+03
30.00	2.387324	8.987742e+03
50.00	3.978874	8.987742e+03

--> |

Variable Browser

Name	Value	Type	Visibility	Memory
H	2	Double	local	216 B
I	50	Double	local	216 B
currents	1x5	Double	local	248 B
electric_f...	1x5	Double	local	248 B
epsilon_0	8.85e-12	Double	local	216 B
i	5	Double	local	216 B
lambda	1e-06	Double	local	216 B
magnetic...	1x5	Double	local	248 B
num_curr...	5	Double	local	216 B

Command History

10/08/2026 14:50:34 -- //

24/08/2026 04:27:36 -- //

24/08/2026 04:29:27 -- //

3*10^-7

6

4

2

5

3

4

5

3

4

5

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Case 3: To analyze the magnetic and electric field intensities due to an infinitely long conductor when there is a potential difference (voltage) across the conductor, simulating a capacitive effect.

Program:

```
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File Edit Format Options Window Execute ?
EXP 12.sci (C:\Users\user\OneDrive\Documents\EXP 12.sci) - SciNotes
EXP 12.scd
5 epsilon_0 = 8.854e-12; // Permittivity of free space (F/m)
6 mu_0 = 4 * pi * 1e-7; // Permeability of free space (H/m)
7 R = 10; // Resistance (ohms)
8
9 // Voltage values
10 V_values = [100, 300, 500, 700, 1000];
11
12 // Distance from conductor
13 H_distance = 1; // Distance in meters
14
15 // Initialize arrays
16 E_values = zeros(1, length(V_values));
17 H_values = zeros(1, length(V_values));
18
19 // Calculate Electric Field and Magnetic Field
20 for i = 1:length(V_values)
21     V = V_values(i);
22
23     // Calculate current using Ohm's law
24     I = V / R;
25
26     // Linear charge density
27     lambda = I;
28
29     // Electric field
30     E_values(i) = lambda / (2 * pi * epsilon_0 * H_distance);
31
32     // Magnetic field intensity
33     H_values(i) = I / (2 * pi * H_distance);
34
35 end
36
37 // Display results
38 disp('Voltage (V) --- Electric Field E (N/C) --- Magnetic Field H (A/m)');
39
40 for i = 1:length(V_values)
41     fprintf('%8.0E --- %18.6e --- %18.6e\n', ...
42           V_values(i), E_values(i), H_values(i));
43 end
44
45
```

Output:

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ex 6.sci

EXP 12.scd

Scilab 2026.1.0 Console

*Voltage (V)	Electric Field E (N/C)	Magnetic Field H (A/m) *
100	1.797548e+11	1.591549
300	5.392645e+11	4.774649
500	8.987742e+11	7.957749
700	1.258284e+12	11.140846
1000	1.797548e+12	15.915494

--> |

Variable Browser

Name	Value	Type	Visibility	Memory
E_values	1x5	Double	local	248 B
H_distance	1	Double	local	216 B
H_values	1x5	Double	local	248 B
I	100	Double	local	216 B
R	10	Double	local	216 B
V	1e+03	Double	local	216 B
V_values	1x5	Double	local	248 B
epsilon_0	8.85e-12	Double	local	216 B
I	5	Double	local	216 B

Command History

```
-- 10/08/2026 14:50:34 -- //
-- 24/08/2026 04:27:36 -- //
-- 24/08/2026 04:29:27 -- //
3*10^-7
6
4
2
5
3
4
5
3
4
5
```

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Result: Thus the program was verified for the magnetic and electric field intensities due to infinitely long conductor.